

Gram Newton–Schulz: A Fast, Hardware-Aware Newton–Schulz Algorithm for Muon

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1 Problem and motivation

- Muon improves hidden-layer optimization by using orthogonalized updates.
- Its main cost is the Newton–Schulz step for approximating the polar factor.
- Gram Newton–Schulz (GNS) moves most work to the smaller Gram matrix and may reduce runtime on rectangular matrices.
- Main project question: can we keep Muon’s quality while reducing orthogonalization cost and maintaining numerical stability?

2 Muon update

$$M_k = \mu M_{k-1} + G_k, \quad W_{k+1} = W_k - \eta \text{polar}(M_k).$$

The polar factor is exact via SVD, but Muon uses a few Newton–Schulz iterations as a practical approximation.

3 Standard Newton–Schulz

Standard NS operates directly on the rectangular matrix X , repeatedly forms $XX^\top$ and applies a fifth-order odd polynomial update. For large aspect ratio $\alpha = m/n$, rectangular GEMMs dominate the cost.

$$\text{standard NS cost} \approx T(3\alpha + 1)n^3 \quad \text{with symmetric GEMMs}$$

4 Project goals

- Reproduce the reported speedup of GNS over standard NS.
- Compare Muon, GramMuon, and AdamW.
- Study fp16 / bf16 stability and the effect of restart.
- Evaluate behavior on different matrix shapes and model scales.

5 Gram Newton–Schulz idea

Instead of iterating on the rectangular matrix X , GNS forms the Gram matrix and updates square matrices only, reducing an approximate inverse square root Q_k :

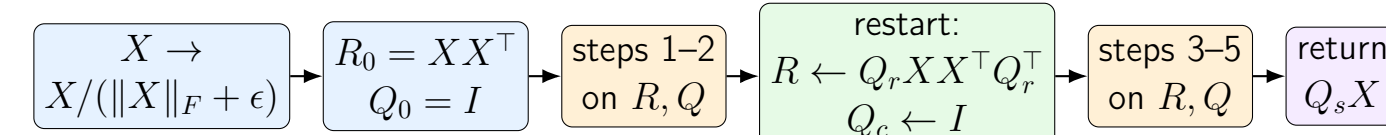
$$R_0 = XX^\top, \quad Q_0 = I, \quad X_T = Q_T X,$$

$$z_t = b_t R_{t-1} + c_t R_{t-1}^2, \quad Q_t = Q_{t-1} + s_t + d_t Q_{t-1}.$$

After T iterations, the orthogonal factor approximation is $Q_T X$.

6 Stabilization by restart

In half precision, the Gram matrix may introduce spurious negative eigenvalues. A restart after iteration r recomputes R from the current approximation.

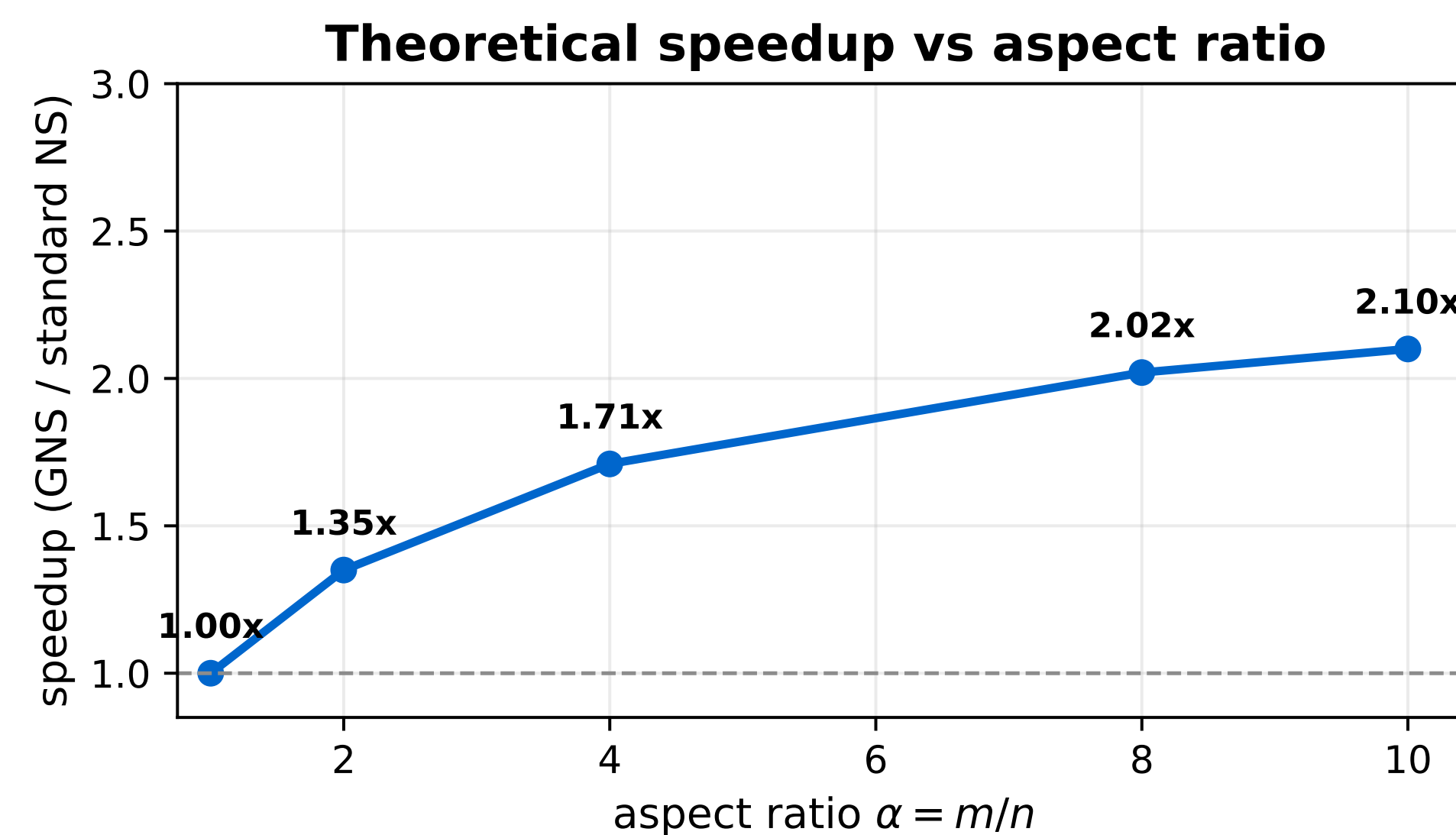


- Restart improves numerical stability.
- It slightly reduces the speed advantage, but remains cheaper than standard NS on wide matrices.

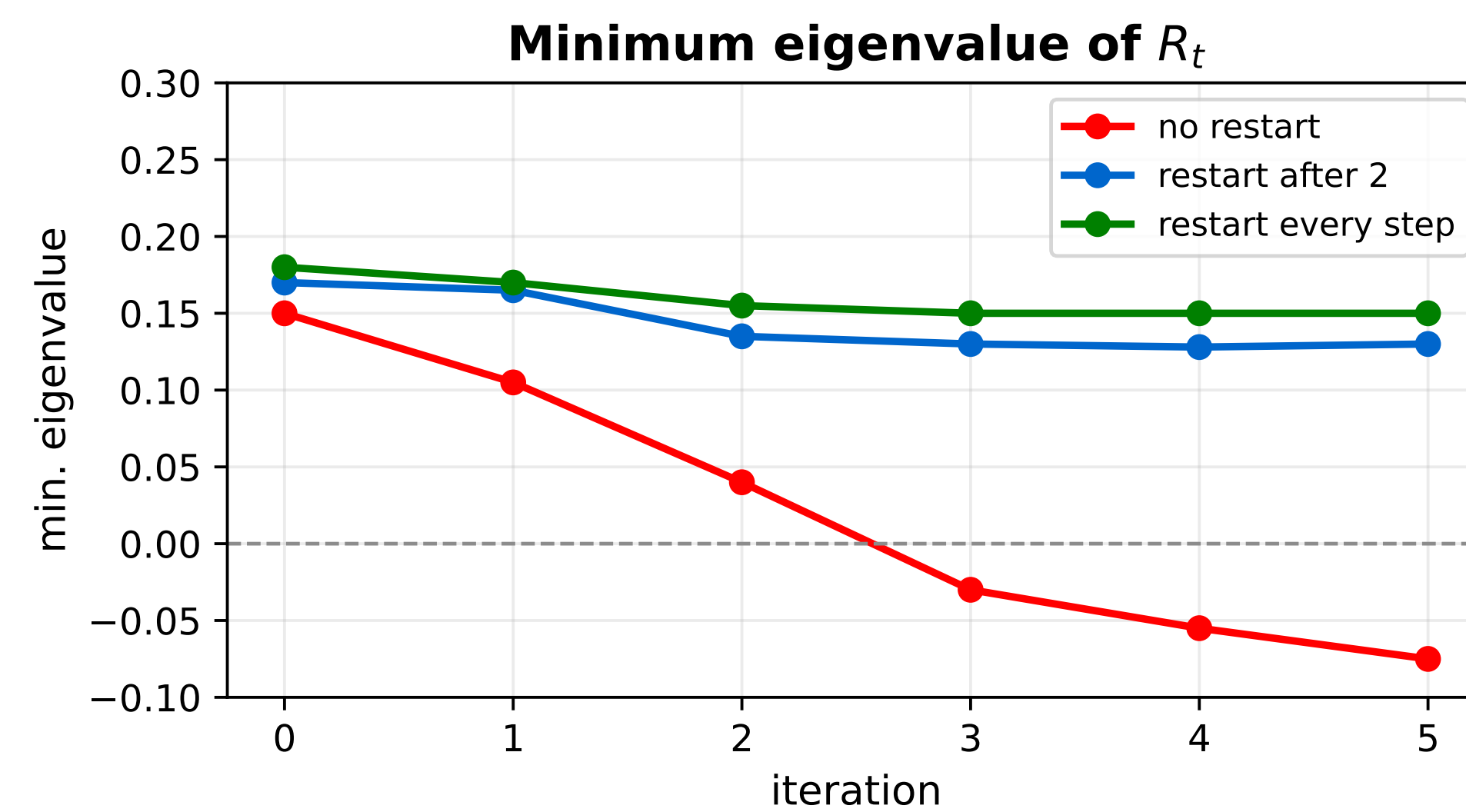
7 Metrics

- validation perplexity / validation loss
- orthogonality defect
- orthogonalization runtime and speedup
- max absolute difference between NS and GNS
- FLOPs
- relative error to exact polar via SVD

8 Theoretical speedup vs aspect ratio

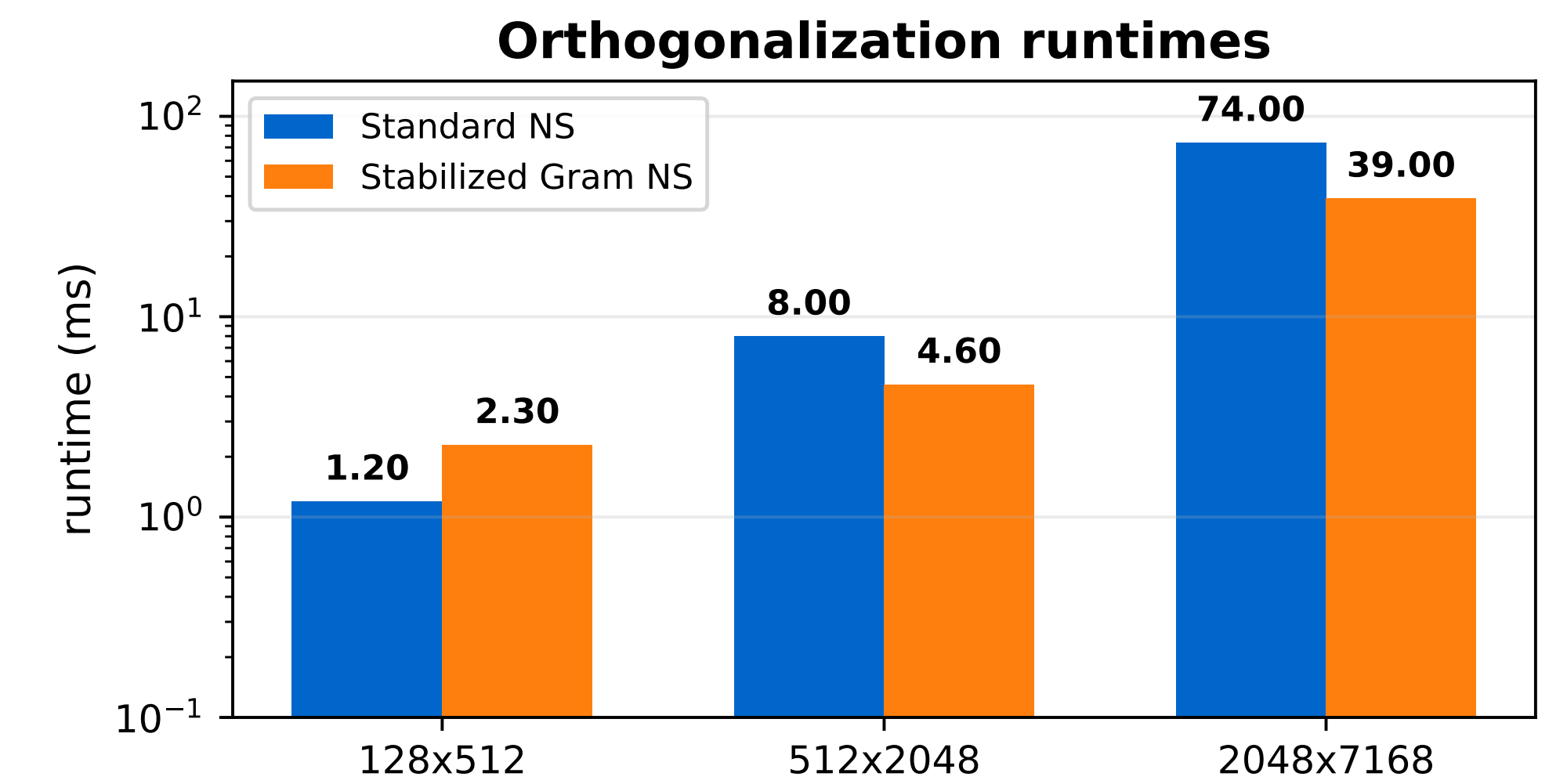


9 Minimum eigenvalue of R_t



10 Experiment 1: orthogonalization benchmark

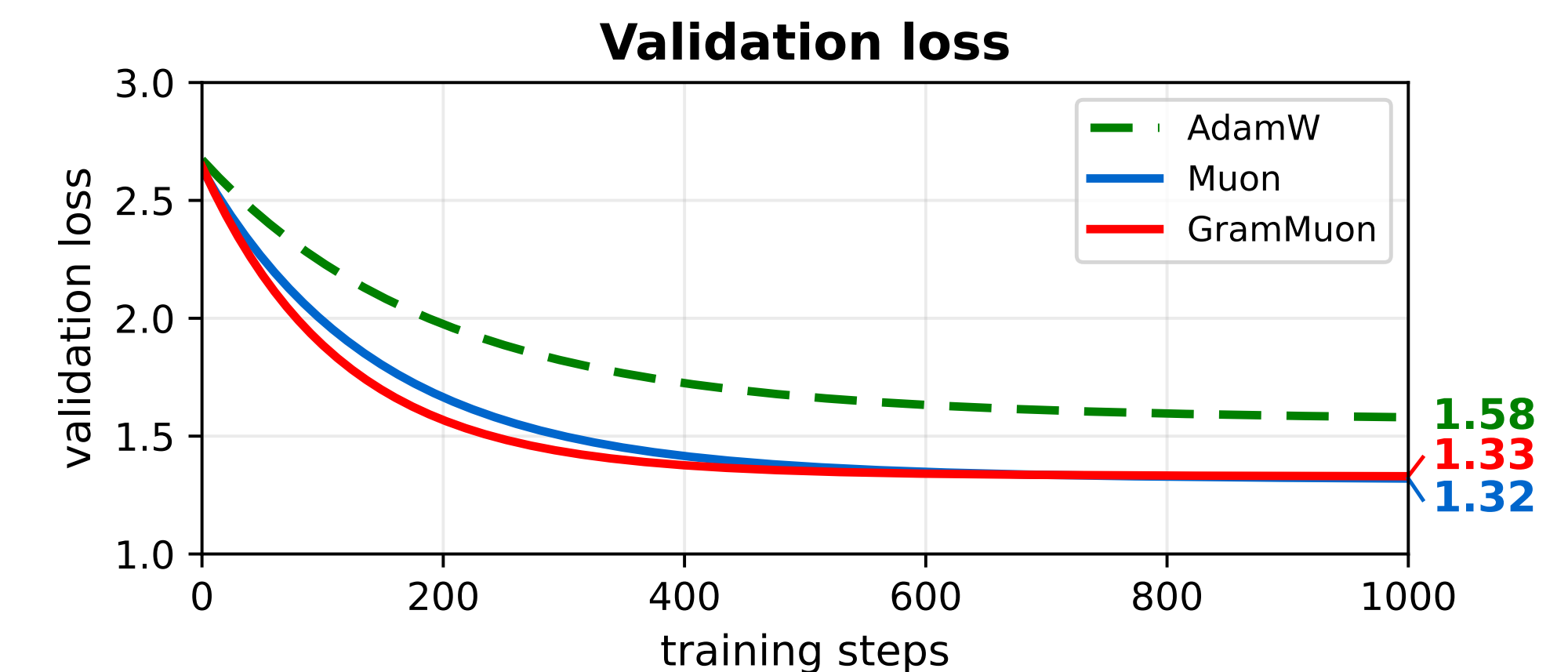
- Compare standard NS and stabilized GNS on random matrices.
- Test fp32 / fp16 / bf16.
- Report time, speedup, orthogonality defect, and max absolute difference.



Trend: GNS may be slower on small matrices but becomes faster on wide large matrices.

11 Experiment 2: optimizer-level quality

- Tiny training proxy: linear regression and a tiny transformer.
- Compare AdamW, Muon, and GramMuon.
- Measure validation loss / perplexity over training steps.



GramMuon matches Muon quality while reducing orthogonalization cost.

12 Experiment 3: restart ablation

- Restart schedules: none, after 2, every step.
- Metrics: failure rate (NaN / inf), orthogonality defect, runtime.

schedule	failure rate	speedup vs standard
none	22%	1.85x
after 2	0%	1.60x
every step	0%	1.18x